

Run: frz150-a0c0e-cl

Log-log loss curves

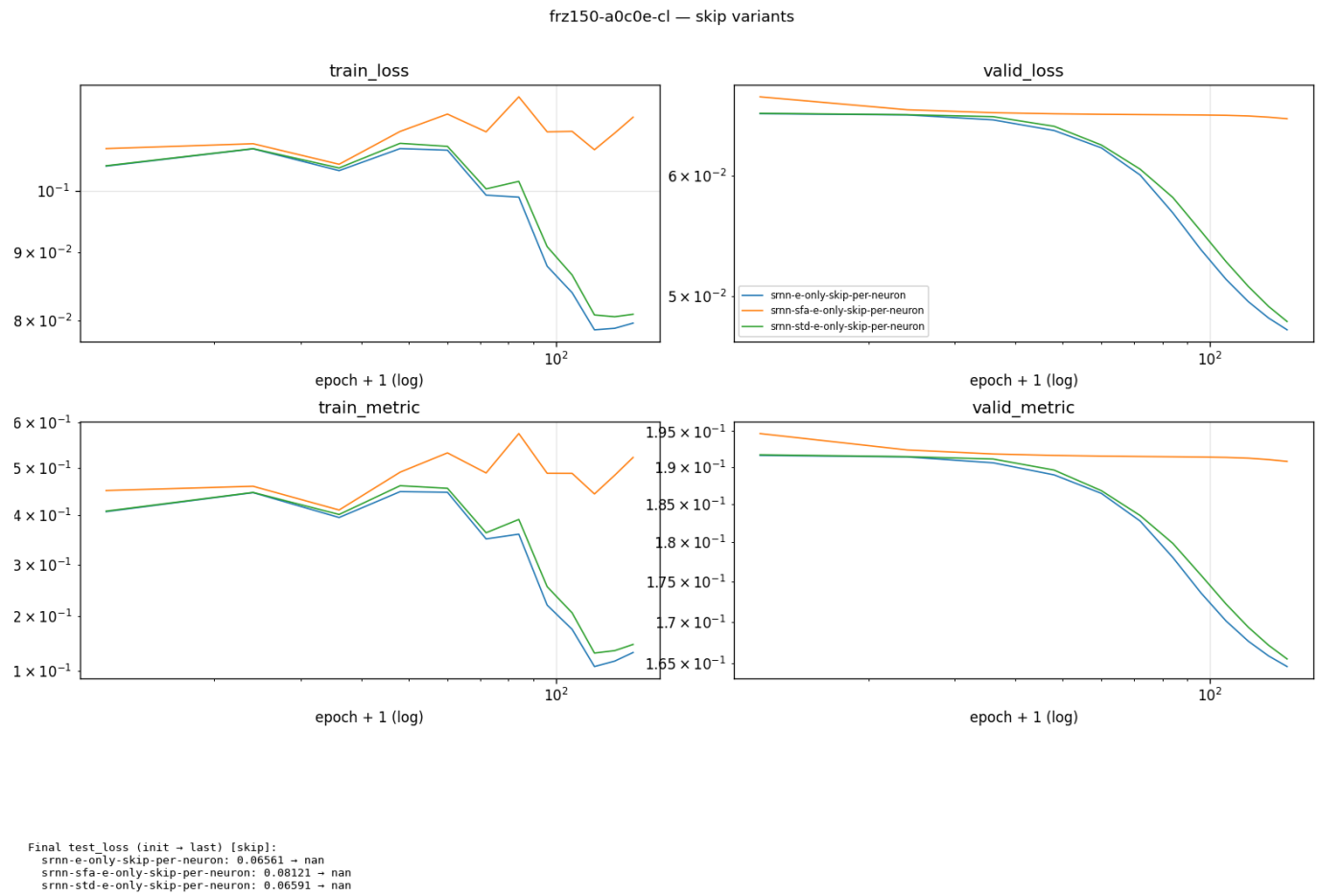
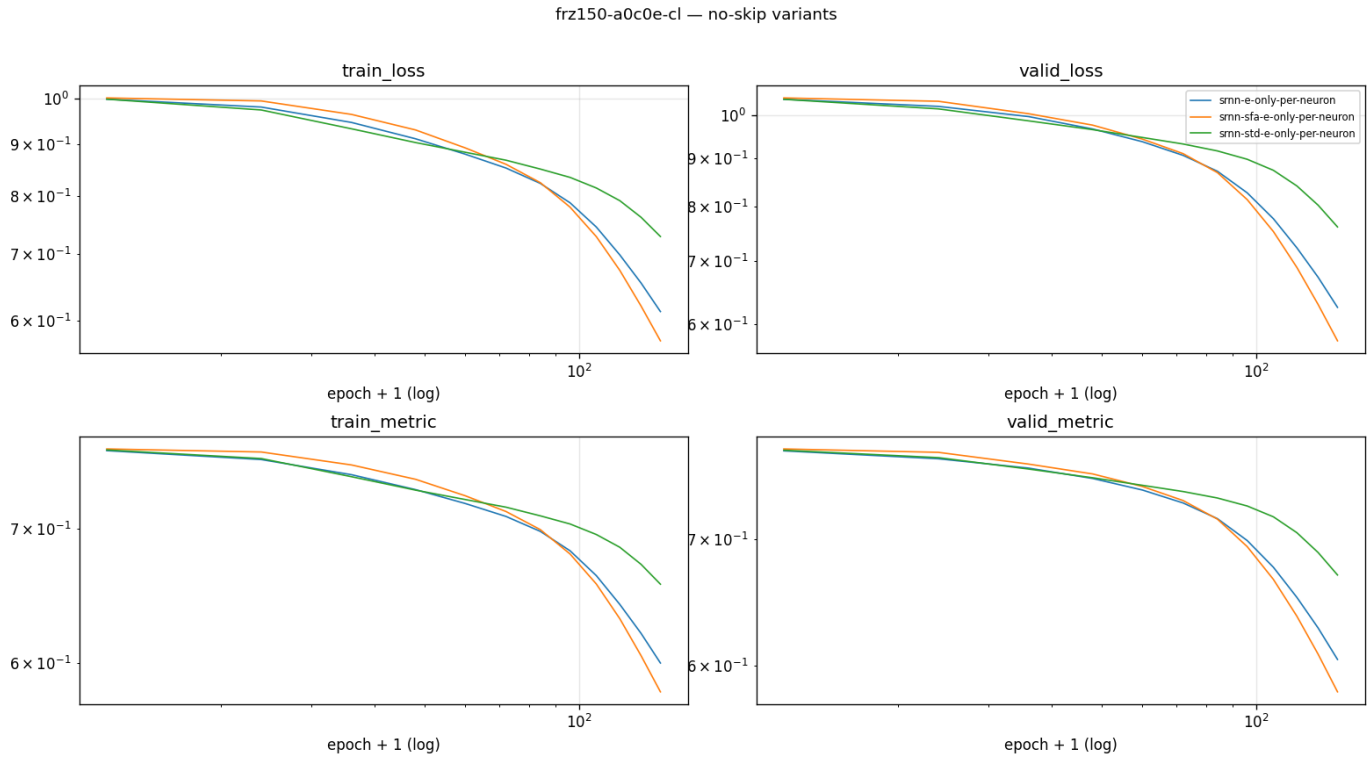


Figure 1: skip variants — log-log loss/metric



Final test loss (init - last) [no-skip]:
 srnn-e-only-per-neuron: 0.98043 - nan
 srnn-sfa-e-only-per-neuron: 0.91505 - nan
 srnn-std-e-only-per-neuron: 0.98077 - nan

Figure 2: no-skip variants — log-log loss/metric

srnn-e-only-per-neuron

frz150-a0c0e-cl — effective time constants (srnn-e-only-per-neuron)

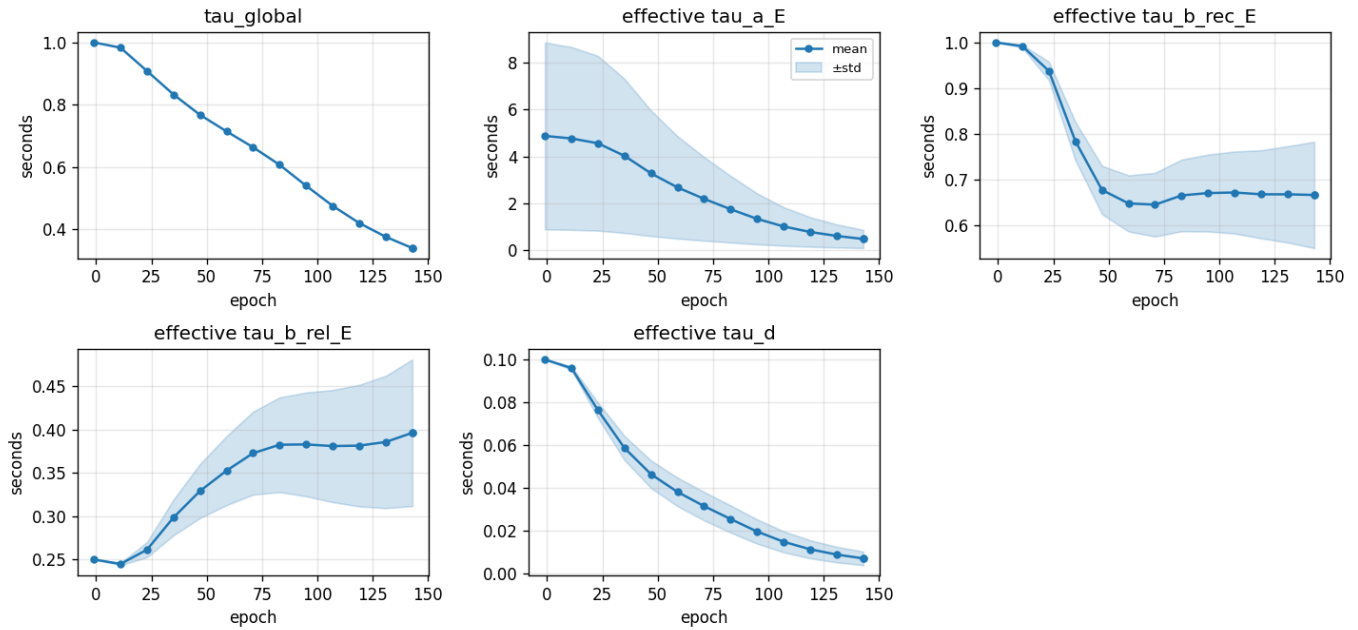
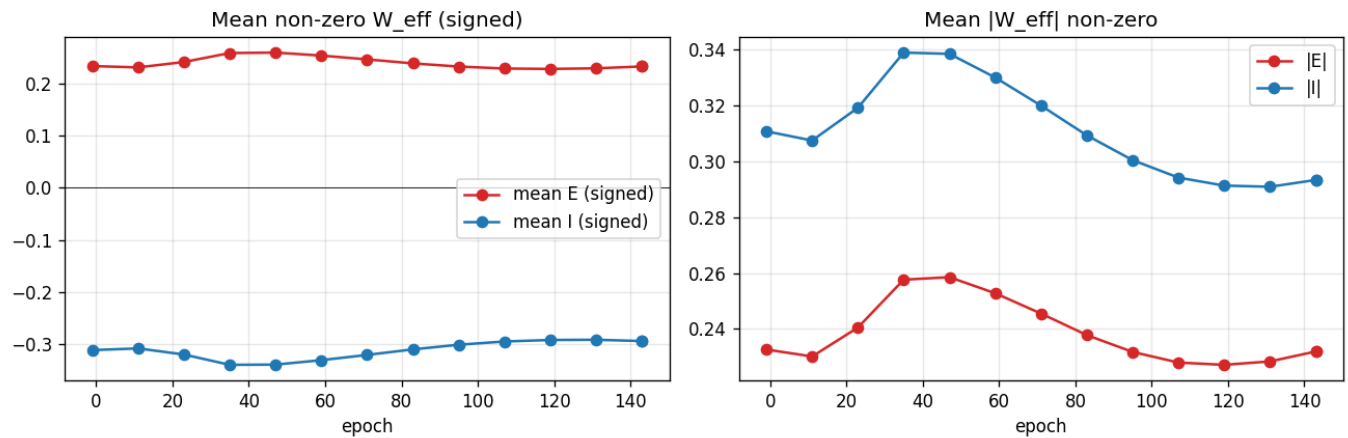


Figure 3: Tau evolution

frz150-a0c0e-cl — recurrent E vs I weights (srnn-e-only-per-neuron)

Figure 4: W_{EI} evolution

srnn-e-only-per-neuron — parameter table

```
# frz150-a0c0e-cl - variant srnn-e-only-per-neuron (k=0)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+0.34039	-	-	scalar
tau_d (s)	+0.1	+0.007162	0	0.003174	300
tau_a_E[j=0] (s)	+0.25	+0.016864	0	0.005739	150
tau_a_E[j=1] (s)	+4.3832	+0.43233	0	0.02709	150
tau_a_E[j=2] (s)	+10	+0.96551	0	0.02078	150
tau_b_rec_E (s)	+1	+0.66645	0	0.1162	150
tau_b_rel_E (s)	+0.25	+0.39631	0	0.08465	150
W_eff (E src, signed)	+0.23267	+0.23196	0.07641	0.0902	15048
W_eff (I src, signed)	-0.31062	-0.29324	0.07724	0.08764	14977
W_eff (E src)	+0.23267	+0.23196	0.07641	0.0902	15048
W_eff (I src)	+0.31062	+0.29324	0.07724	0.08764	14977
W_in_eff (input neurons)	-0.0003386	+0.0012009	0.1012	0.06848	6675
c_E[j=0] (SFA coupling)	+0.05	+0.026238	0	0.0128	150
c_E[j=1] (SFA coupling)	+0.05	+0.018407	0	0.004744	150
c_E[j=2] (SFA coupling)	+0.05	+0.018962	0	0.004853	150
c_0_E[j=0] (SFA offset)	+0	+0	0	0	150
c_0_E[j=1] (SFA offset)	+0	+0	0	0	150
c_0_E[j=2] (SFA offset)	+0	+0	0	0	150
a_0 (threshold)	+0.35	+0.35	0	0	300
readout_weight	-2.8854e-05	-0.0023459	0.06604	0.2778	6675
readout_bias	+0	-0.027527	0	0.1935	89
ic.ic (per-variant init)	+0.10386	+0.10386	0.3828	0.3828	900

srnn-e-only-skip-per-neuron

frz150-a0c0e-cl — effective time constants (srnn-e-only-skip-per-neuron)

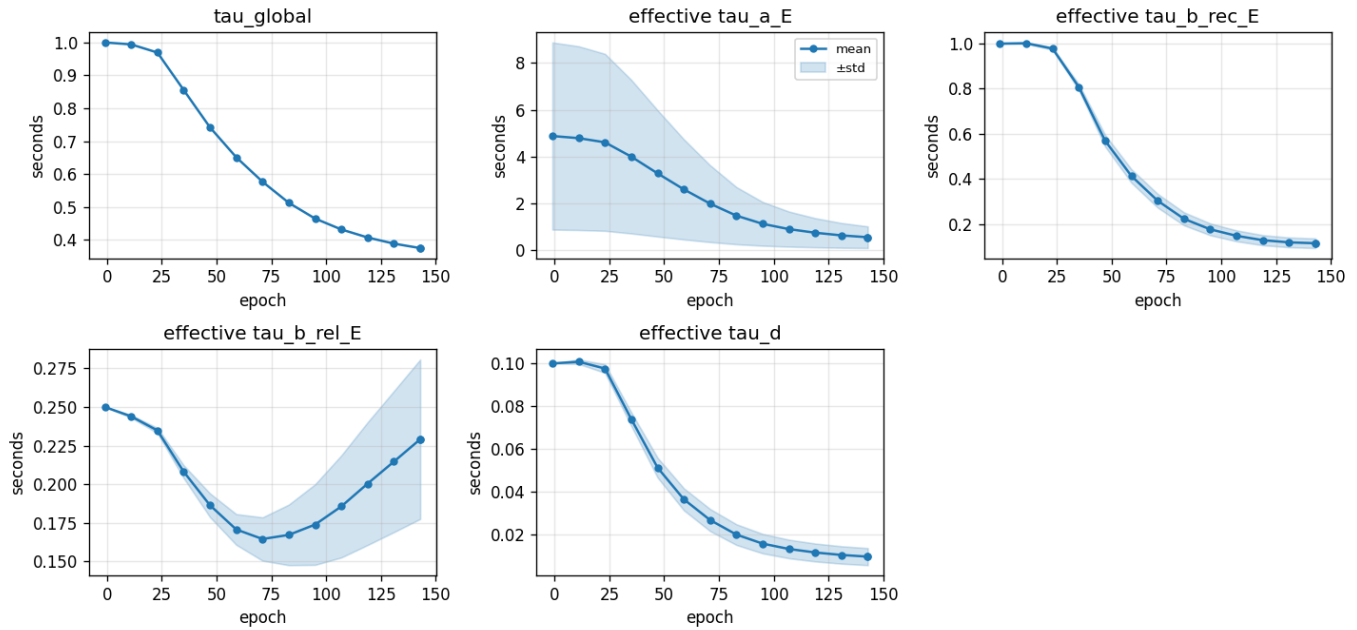
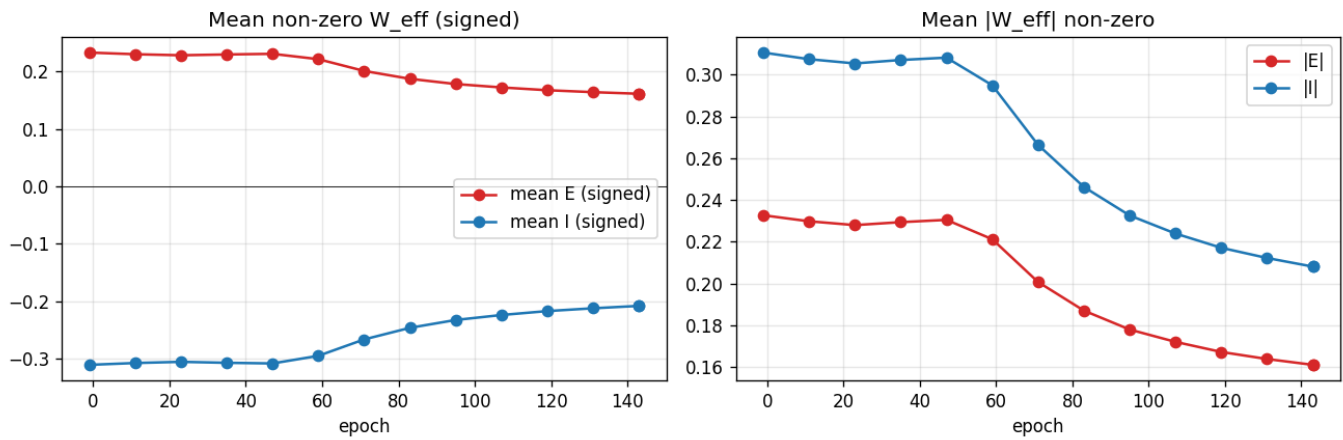


Figure 5: Tau evolution

frz150-a0c0e-cl — recurrent E vs I weights (srnn-e-only-skip-per-neuron)

Figure 6: W_{EI} evolution

srnn-e-only-skip-per-neuron — parameter table

frz150-a0c0e-cl - variant srnn-e-only-skip-per-neuron (k=1)
Effective values (post-transform). Means/stds computed over the indicated dimension.

effective parameter	init mean	final mean	init std	final std	n
tau_global (s)	+1	+0.37475	-	-	scalar
tau_d (s)	+0.1	+0.0093965	0	0.004008	300
tau_a_E[j=0] (s)	+0.25	+0.018205	0	0.0071	150
tau_a_E[j=1] (s)	+4.3832	+0.50243	0	0.01512	150
tau_a_E[j=2] (s)	+10	+1.1475	0	0.006022	150
tau_b_rec_E (s)	+1	+0.1144	0	0.0215	150
tau_b_rel_E (s)	+0.25	+0.2291	0	0.05166	150
W_eff (E src, signed)	+0.23267	+0.16102	0.07641	0.06004	15048
W_eff (I src, signed)	-0.31062	-0.20807	0.07724	0.06089	14977
W_eff (E src)	+0.23267	+0.16102	0.07641	0.06004	15048
W_eff (I src)	+0.31062	+0.20807	0.07724	0.06089	14977
W_in_eff (input neurons)	-0.00084237	-0.00021733	0.1008	0.02974	6675
c_E[j=0] (SFA coupling)	+0.05	+0.064445	0	0.02727	150
c_E[j=1] (SFA coupling)	+0.05	+0.038848	0	0.009048	150
c_E[j=2] (SFA coupling)	+0.05	+0.039301	0	0.00884	150
c_0_E[j=0] (SFA offset)	+0	+0	0	0	150
c_0_E[j=1] (SFA offset)	+0	+0	0	0	150
c_0_E[j=2] (SFA offset)	+0	+0	0	0	150
a_0 (threshold)	+0.35	+0.35	0	0	300
readout_weight	+0.00078219	-0.0041568	0.06654	0.1669	6675
readout_bias	+0	+0.075051	0	0.09914	89
ic.ic (per-variant init)	+0.10327	+0.10327	0.3829	0.3829	900

srnn-sfa-e-only-per-neuron

frz150-a0c0e-cl — effective time constants (srnn-sfa-e-only-per-neuron)

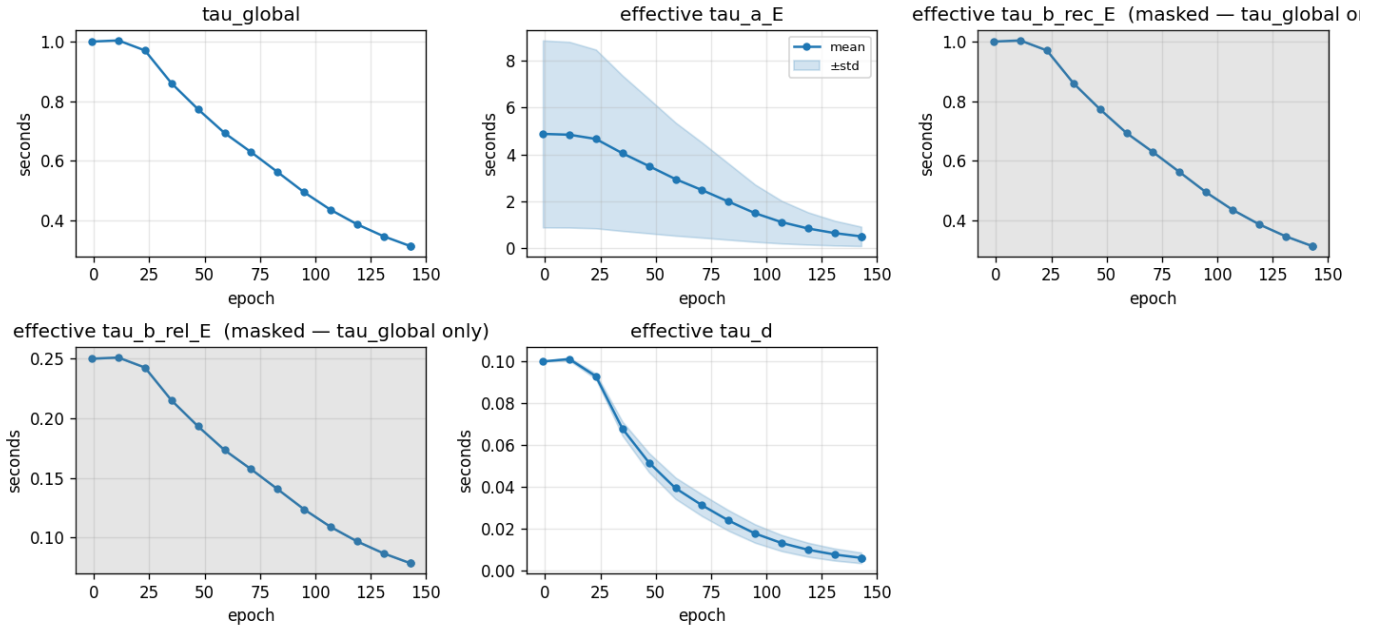


Figure 7: Tau evolution

frz150-a0c0e-cl — recurrent E vs I weights (srnn-sfa-e-only-per-neuron)

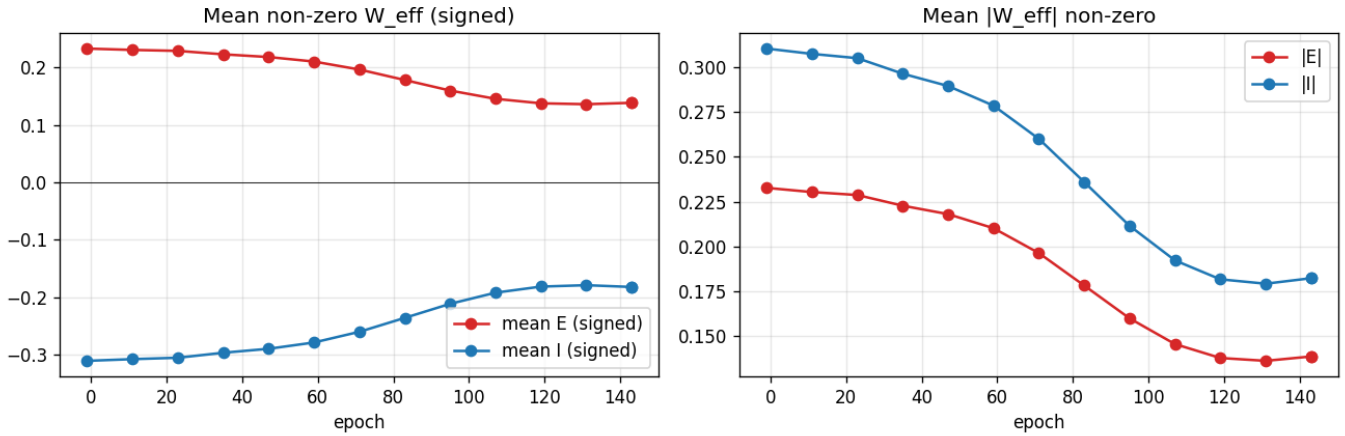


Figure 8: W_EI evolution

srnn-sfa-e-only-per-neuron — parameter table

frz150-a0c0e-cl - variant srnn-sfa-e-only-per-neuron (k=2)
Effective values (post-transform). Means/stds computed over the indicated dimension.

effective parameter	init mean	final mean	init std	final std	n
tau_global (s)	+1	+0.31358	-	-	scalar
tau_d (s)	+0.1	+0.0061526	0	0.002533	300
tau_a_E[j=0] (s)	+0.25	+0.01798	0	0.005394	150
tau_a_E[j=1] (s)	+4.3832	+0.4452	0	0.02597	150
tau_a_E[j=2] (s)	+10	+1.0277	0	0.0222	150
tau_b_rec_E (s)	+1	+0.31358	0	0	150
tau_b_rel_E (s)	+0.25	+0.078396	0	0	150
W_eff (E src, signed)	+0.23267	+0.13831	0.07641	0.04941	15048
W_eff (I src, signed)	-0.31062	-0.18211	0.07724	0.05093	14977
W_eff (E src)	+0.23267	+0.13831	0.07641	0.04941	15048
W_eff (I src)	+0.31062	+0.18211	0.07724	0.05093	14977
W_in_eff (input neurons)	-0.00015382	-0.0014175	0.09934	0.05862	6675
c_E[j=0] (SFA coupling)	+0.05	+0.059952	0	0.01667	150
c_E[j=1] (SFA coupling)	+0.05	+0.044852	0	0.007163	150
c_E[j=2] (SFA coupling)	+0.05	+0.044457	0	0.007396	150
c_0_E[j=0] (SFA offset)	+0	+0	0	0	150
c_0_E[j=1] (SFA offset)	+0	+0	0	0	150
c_0_E[j=2] (SFA offset)	+0	+0	0	0	150
a_0 (threshold)	+0.35	+0.35	0	0	300
readout_weight	-0.00098972	+0.0075209	0.06678	0.2081	6675
readout_bias	+0	-0.06755	0	0.1037	89
ic.ic (per-variant init)	+0.13078	+0.13078	0.4172	0.4172	900

srnn-sfa-e-only-skip-per-neuron

frz150-a0c0e-cl — effective time constants (srnn-sfa-e-only-skip-per-neuron)

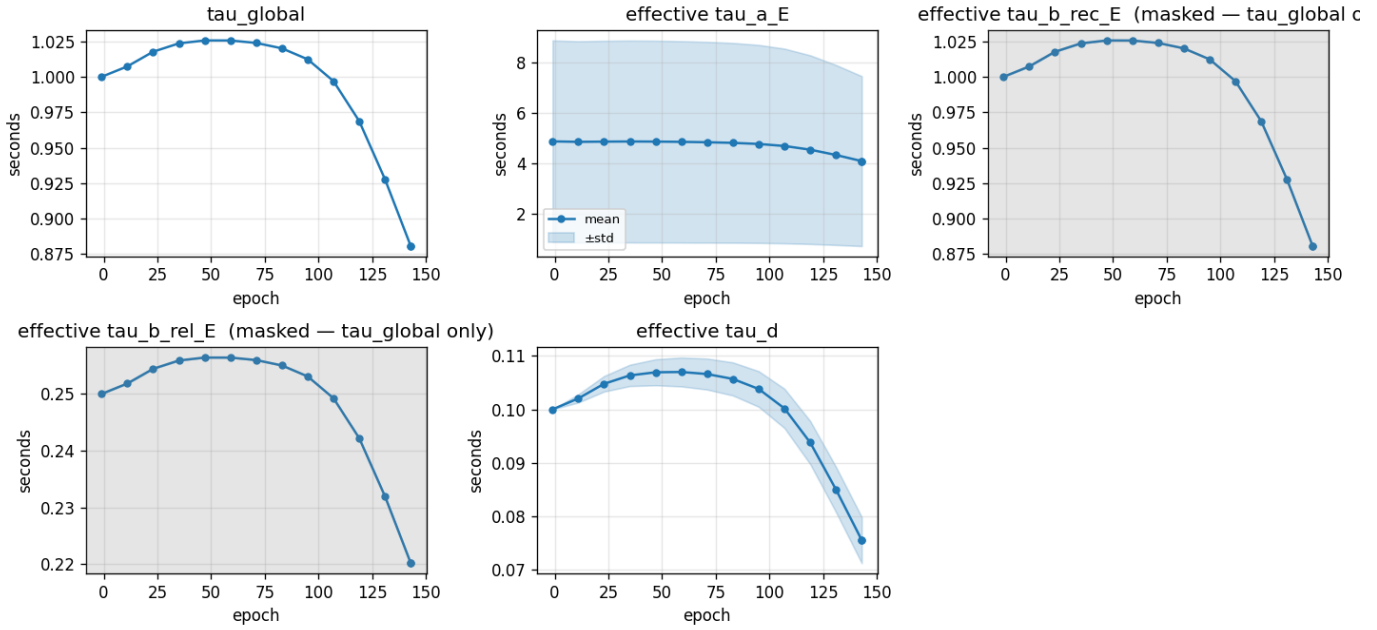
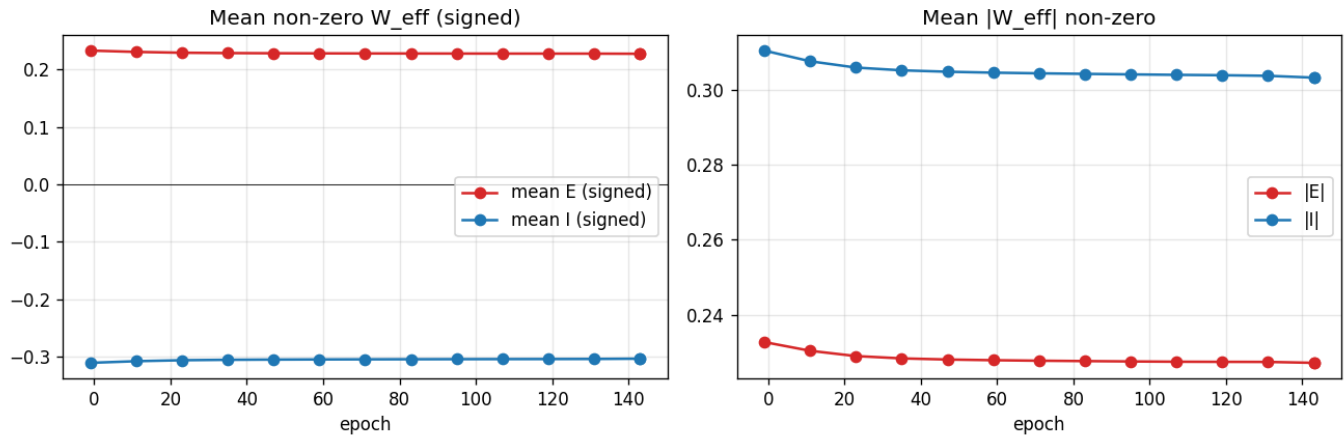


Figure 9: Tau evolution

frz150-a0c0e-cl — recurrent E vs I weights (srnn-sfa-e-only-skip-per-neuron)

Figure 10: W_{EI} evolution

srnn-sfa-e-only-skip-per-neuron — parameter table

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# frz150-a0c0e-cl - variant srnn-sfa-e-only-skip-per-neuron (k=3)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+0.88066	-	-	scalar
tau_d (s)	+0.1	+0.07556	0	0.004313	300
tau_a_E[j=0] (s)	+0.25	+0.20439	0	0.01152	150
tau_a_E[j=1] (s)	+4.3832	+3.6736	0	0.03808	150
tau_a_E[j=2] (s)	+10	+8.4115	0	0.01464	150
tau_b_rec_E (s)	+1	+0.88066	0	0	150
tau_b_rel_E (s)	+0.25	+0.22016	0	0	150
W_eff (E src, signed)	+0.23267	+0.22712	0.07641	0.07492	15048
W_eff (I src, signed)	-0.31062	-0.30339	0.07724	0.0758	14977
W_eff (E src)	+0.23267	+0.22712	0.07641	0.07492	15048
W_eff (I src)	+0.31062	+0.30339	0.07724	0.0758	14977
W_in_eff (input neurons)	-0.0011621	-0.00093949	0.09962	0.09398	6675
c_E[j=0] (SFA coupling)	+0.05	+0.052804	0	0.00683	150
c_E[j=1] (SFA coupling)	+0.05	+0.05201	0	0.0008557	150
c_E[j=2] (SFA coupling)	+0.05	+0.051995	0	0.001086	150
c_0_E[j=0] (SFA offset)	+0	+0	0	0	150
c_0_E[j=1] (SFA offset)	+0	+0	0	0	150
c_0_E[j=2] (SFA offset)	+0	+0	0	0	150
a_0 (threshold)	+0.35	+0.35	0	0	300
readout_weight	+0.00016801	-0.0043514	0.0673	0.06344	6675
readout_bias	+0	+0.0042362	0	0.008181	89
ic.ic (per-variant init)	+0.09231	+0.09231	0.5294	0.5294	900

srnn-std-e-only-per-neuron

frz150-a0c0e-cl — effective time constants (srnn-std-e-only-per-neuron)

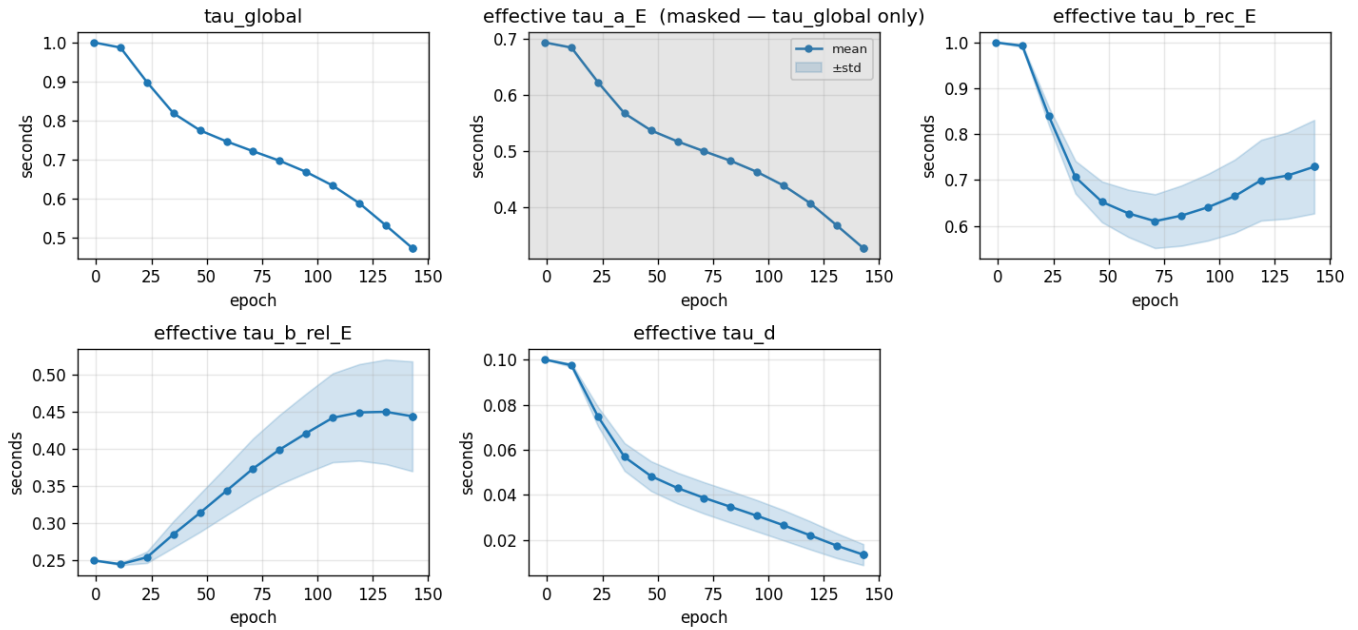
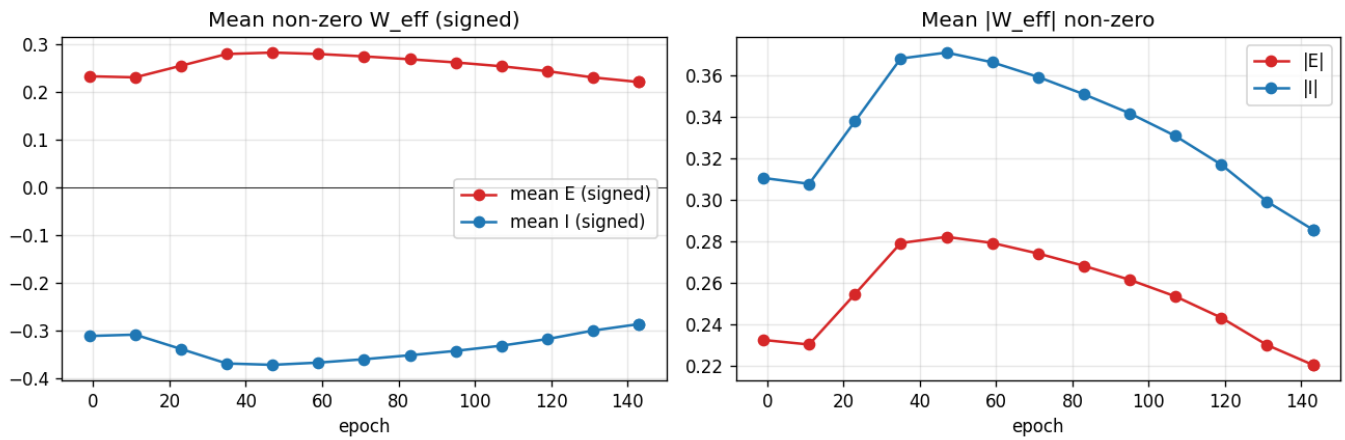


Figure 11: Tau evolution

frz150-a0c0e-cl — recurrent E vs I weights (srnn-std-e-only-per-neuron)

Figure 12: W_{EI} evolution

srnn-std-e-only-per-neuron — parameter table

```
# frz150-a0c0e-cl - variant srnn-std-e-only-per-neuron (k=4)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+0.47323	-	-	scalar
tau_d (s)	+0.1	+0.013463	0	0.00467	300
tau_b_rec_E (s)	+1	+0.72903	0	0.1019	150
tau_b_rel_E (s)	+0.25	+0.44383	0	0.07382	150
W_eff (E src, signed)	+0.23267	+0.22068	0.07641	0.07992	15048
W_eff (I src, signed)	-0.31062	-0.28579	0.07724	0.08107	14977
W_eff (E src)	+0.23267	+0.22068	0.07641	0.07992	15048
W_eff (I src)	+0.31062	+0.28579	0.07724	0.08107	14977
W_in_eff (input neurons)	+0.00019944	-0.0012099	0.09994	0.09731	6675
a_0 (threshold)	+0.35	+0.35	0	0	300
readout_weight	+0.00074661	+0.0073141	0.06657	0.2248	6675
readout_bias	+0	-0.050919	0	0.2471	89
ic.ic (per-variant init)	+0.090819	+0.090819	0.3835	0.3835	900

srnn-std-e-only-skip-per-neuron

frz150-a0c0e-cl — effective time constants (srnn-std-e-only-skip-per-neuron)

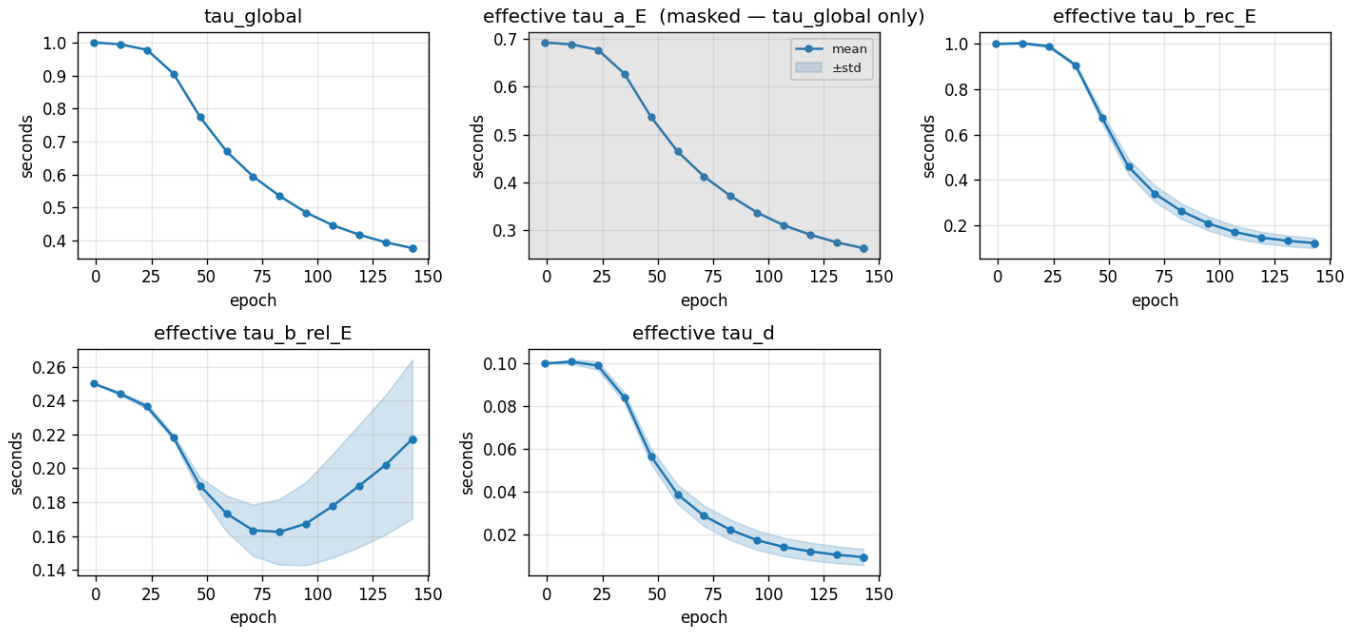


Figure 13: Tau evolution

frz150-a0c0e-cl — recurrent E vs I weights (srnn-std-e-only-skip-per-neuron)

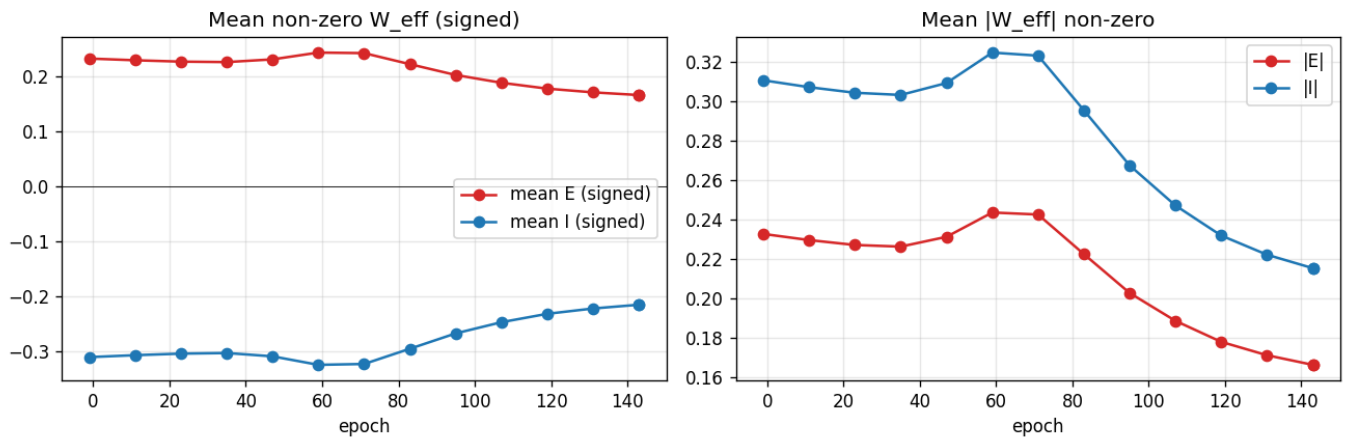


Figure 14: W_EI evolution

srnn-std-e-only-skip-per-neuron — parameter table

```
# frz150-a0c0e-cl - variant srnn-std-e-only-skip-per-neuron (k=5)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+0.37754	-	-	scalar
tau_d (s)	+0.1	+0.0096444	0	0.003793	300
tau_b_rec_E (s)	+1	+0.12067	0	0.02221	150
tau_b_rel_E (s)	+0.25	+0.21729	0	0.04691	150
W_eff (E src, signed)	+0.23267	+0.16627	0.07641	0.06157	15048
W_eff (I src, signed)	-0.31062	-0.2153	0.07724	0.06187	14977
W_eff (E src)	+0.23267	+0.16627	0.07641	0.06157	15048
W_eff (I src)	+0.31062	+0.2153	0.07724	0.06187	14977
W_in_eff (input neurons)	+0.00036119	+0.00041135	0.09956	0.03142	6675
a_0 (threshold)	+0.35	+0.35	0	0	300
readout_weight	+0.00055372	-0.0028452	0.06572	0.1506	6675
readout_bias	+0	+0.0040465	0	0.06836	89
ic.ic (per-variant init)	+0.083543	+0.083543	0.384	0.384	900